

ECEN 629 Convex Optimization for Electrical Engineering

Chapter 7. Statistical estimation

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Parametric distribution estimation

- distribution estimation problem: estimate probability density $p(y)$ of a random variable from observed values
- parametric distribution estimation: choose from a family of densities $p_x(y)$, indexed by a parameter x

maximum likelihood estimation

$$\text{maximize (over } x) \quad \log p_x(y)$$

- y is observed value
- $l(x) = \log p_x(y)$ is called log-likelihood function
- can add constraints $x \in C$ explicitly, or define $p_x(y) = 0$ for $x \notin C$
- a convex optimization problem if $\log p_x(y)$ is concave in x for fixed y

Linear measurements with IID noise

linear measurement model

$$y_i = a_i^T x + v_i, \quad i = 1, \dots, m$$

- $x \in \mathbf{R}^n$ is vector of unknown parameters
- v_i is IID measurement noise, with density $p(z)$
- y_i is measurement: $y \in \mathbf{R}^m$ has density $p_x(y) = \prod_{i=1}^m p(y_i - a_i^T x)$

maximum likelihood estimate: any solution x of

$$\text{maximize } l(x) = \sum_{i=1}^m \log p(y_i - a_i^T x)$$

(y is observed value)

examples

- Gaussian noise $\mathcal{N}(0, \sigma^2)$: $p(z) = (2\pi\sigma^2)^{-1/2} e^{-z^2/(2\sigma^2)}$,

$$l(x) = -\frac{m}{2} \log(2\pi\sigma^2) - \frac{1}{2\sigma^2} \sum_{i=1}^m (a_i^T x - y_i)^2$$

ML estimate is LS solution

- Laplacian noise: $p(z) = (1/(2a)) e^{-|z|/a}$,

$$l(x) = -m \log(2a) - \frac{1}{a} \sum_{i=1}^m |a_i^T x - y_i|$$

ML estimate is ℓ_1 -norm solution

- uniform noise on $[-a, a]$:

$$l(x) = \begin{cases} -m \log(2a) & |a_i^T x - y_i| \leq a, \quad i = 1, \dots, m \\ -\infty & \text{otherwise} \end{cases}$$

ML estimate is any x with $|a_i^T x - y_i| \leq a$

Exercise: Linear Measurement with IID Truncated Gaussian Noise

Suppose the measurement noise has the truncated Gaussian noise distribution

$$p_x(y) = \frac{2}{\sqrt{2\pi\sigma^2}} \exp \left[-\frac{(y-x)^2}{2\sigma^2} \right], \quad y \geq x,$$

and $p_x(y) = 0$ otherwise.

1. Write down the ML estimation problem using m iid scalar samples $\{y_i\}$ as an optimization problem;
2. What is the variable? Is this a convex optimization problem?
3. Find an closed-form solution for the problem.

Exercise': Linear Measurement with IID Truncated Gaussian Noise

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$$p_x(y) = \frac{2}{\sqrt{2\pi\sigma^2}} \exp \left[-\frac{(y-x)^2}{2\sigma^2} \right], \quad y \geq x - 5,$$

and $p_x(y) = 0$ otherwise.

1. Write down the ML estimation problem using m iid scalar samples $\{y_i\}$ as an optimization problem;
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Logistic regression

random variable $y \in \{0, 1\}$ with distribution

$$p = \mathbf{prob}(y = 1) = \frac{\exp(a^T u + b)}{1 + \exp(a^T u + b)}$$

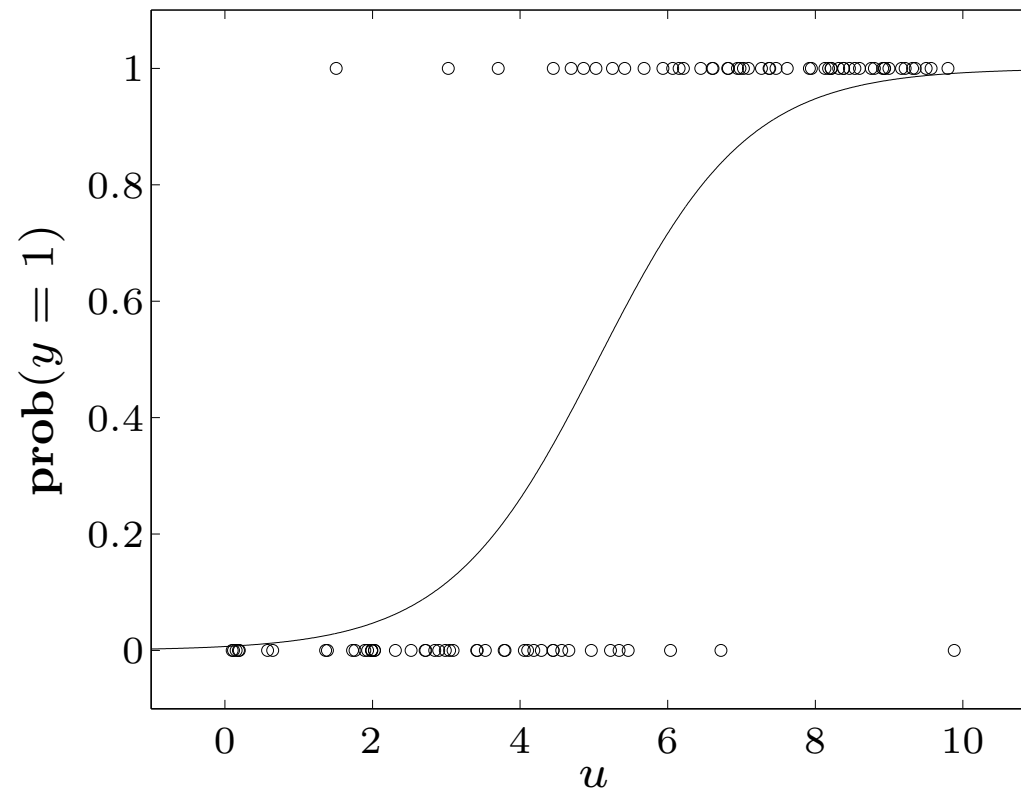
- a, b are parameters; $u \in \mathbf{R}^n$ are (observable) explanatory variables
- estimation problem: estimate a, b from m observations (u_i, y_i)

log-likelihood function (for $y_1 = \dots = y_k = 1, y_{k+1} = \dots = y_m = 0$):

$$\begin{aligned} l(a, b) &= \log \left(\prod_{i=1}^k \frac{\exp(a^T u_i + b)}{1 + \exp(a^T u_i + b)} \prod_{i=k+1}^m \frac{1}{1 + \exp(a^T u_i + b)} \right) \\ &= \sum_{i=1}^k (a^T u_i + b) - \sum_{i=1}^m \log(1 + \exp(a^T u_i + b)) \end{aligned}$$

concave in a, b

example ($n = 1$, $m = 50$ measurements)



- circles show 50 points (u_i, y_i)
- solid curve is ML estimate of $p = \exp(au + b) / (1 + \exp(au + b))$

(Binary) hypothesis testing

detection (hypothesis testing) problem

given observation of a random variable $X \in \{1, \dots, n\}$, choose between:

- hypothesis 1: X was generated by distribution $p = (p_1, \dots, p_n)$
- hypothesis 2: X was generated by distribution $q = (q_1, \dots, q_n)$

randomized detector

- a nonnegative matrix $T \in \mathbf{R}^{2 \times n}$, with $\mathbf{1}^T T = \mathbf{1}^T$
- if we observe $X = k$, we choose hypothesis 1 with probability t_{1k} , hypothesis 2 with probability t_{2k}
- if all elements of T are 0 or 1, it is called a deterministic detector

detection probability matrix:

$$D = \begin{bmatrix} Tp & Tq \end{bmatrix} = \begin{bmatrix} 1 - P_{fp} & P_{fn} \\ P_{fp} & 1 - P_{fn} \end{bmatrix}$$

- P_{fp} is probability of selecting hypothesis 2 if X is generated by distribution 1 (false positive)
- P_{fn} is probability of selecting hypothesis 1 if X is generated by distribution 2 (false negative)

multicriterion formulation of detector design

$$\begin{array}{ll} \text{minimize (w.r.t. } \mathbf{R}_+^2) & (P_{fp}, P_{fn}) = ((Tp)_2, (Tq)_1) \\ \text{subject to} & t_{1k} + t_{2k} = 1, \quad k = 1, \dots, n \\ & t_{ik} \geq 0, \quad i = 1, 2, \quad k = 1, \dots, n \end{array}$$

variable $T \in \mathbf{R}^{2 \times n}$

scalarization (with weight $\lambda > 0$)

$$\begin{array}{ll}\text{minimize} & (Tp)_2 + \lambda(Tq)_1 \\ \text{subject to} & t_{1k} + t_{2k} = 1, \quad t_{ik} \geq 0, \quad i = 1, 2, \quad k = 1, \dots, n\end{array}$$

an LP with a simple analytical solution

$$(t_{1k}, t_{2k}) = \begin{cases} (1, 0) & p_k \geq \lambda q_k \\ (0, 1) & p_k < \lambda q_k \end{cases}$$

- a deterministic detector, given by a likelihood ratio test
- if $p_k = \lambda q_k$ for some k , any value $0 \leq t_{1k} \leq 1$, $t_{1k} = 1 - t_{2k}$ is optimal (*i.e.*, Pareto-optimal detectors include non-deterministic detectors)

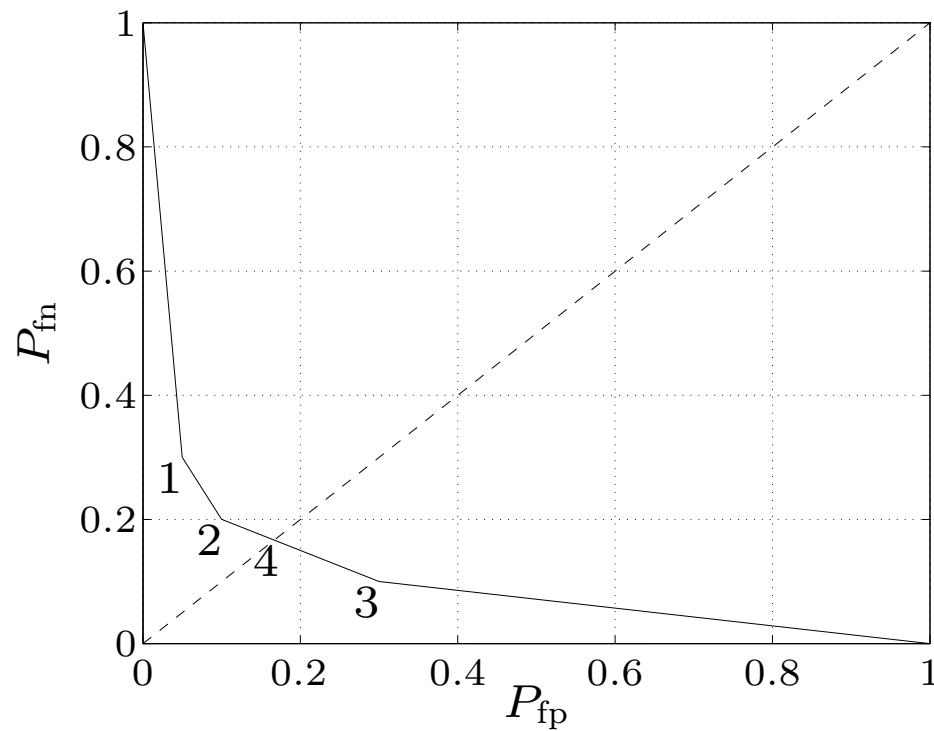
minimax detector

$$\begin{array}{ll}\text{minimize} & \max\{P_{\text{fp}}, P_{\text{fn}}\} = \max\{(Tp)_2, (Tq)_1\} \\ \text{subject to} & t_{1k} + t_{2k} = 1, \quad t_{ik} \geq 0, \quad i = 1, 2, \quad k = 1, \dots, n\end{array}$$

an LP; solution is usually not deterministic

example

$$P = \begin{bmatrix} 0.70 & 0.10 \\ 0.20 & 0.10 \\ 0.05 & 0.70 \\ 0.05 & 0.10 \end{bmatrix}$$



solutions 1, 2, 3 (and endpoints) are deterministic; 4 is minimax detector

Generalization to Non-binary Detection

The probability of m hypotheses: $P = (p_1, p_2, \dots, p_m)$

- The detection matrix $D = T \cdot P$;
- $P_i^d = D_{ii} = \mathbf{Prob}(\hat{\theta} = i | \theta = i)$;
- $P_i^e = 1 - D_{ii} = 1 - \mathbf{Prob}(\hat{\theta} = i | \theta = i)$;
- Placing various constraints on $D_{i,j}$: e.g.,
 - $D_{ii} \geq L_{ii}$ for all i ;
 - $D_{ij} \leq U_{i,j}$ for all $i \neq j$.

Linear constraints in general: type of optimization problem depends on the objective function.